Human Fetal Heart PCW 11–13

Multi-modal Single-cell and Spatial Atlas — Data Overview

PCW12_analysis project

April 30, 2026

Abstract

This document summarizes the two human fetal heart datasets that underlie the PCW12_analysis project: a 95,684-cell scRNA-seq atlas spanning post-conceptional weeks (PCW) 11–13 across 8 donors, and a 100,000-cell MERFISH 3D spatial transcriptomics dataset of a single PCW-12 specimen (downsampled from a 1.7M-cell full atlas) with a 238-gene panel and 53 serial sections. We characterize per-cell quality control, sample composition, cell-type abundance in both modalities, and the spatial layout of the MERFISH specimen. A cross-modality comparison shows strong concordance in broad cell-type proportions, supporting the use of these datasets together for downstream tasks such as gene imputation and disease-gene spatial mapping.

1 Datasets

Property	**scRNA-seq**	**MERFISH (3D spatial)**
File	`all_healthy_RoundedPCW11-13.h5ad`	`full_heart_final_aug2025_update_downsampled`
Cells	95,684	100,000 (subsampled from ∼1.7M)
Genes / panel size	31,019 (full transcriptome)	238 (targeted MERFISH panel)
Developmental stage	PCW 11, 12, 13	PCW 12
Donors / specimens	8 donors, 24 samples	1 specimen, 53 serial sections
Modality batches	—	2 (ATR: atrial, VEN: ventricular)
Cell-type annotations	11 broad / 111 fine	34 fine
Layers	`counts` (raw)	`counts` (raw); X is log1p
Embeddings shipped	—	`X_pca`, `X_pca_harmony`, `X_umap`, `X_spatial` (2D), `X_spateo_update` (3D)

Table 1. High-level summary of the two h5ad inputs used in this project.

The scRNA-seq atlas is multi-donor and spans PCW 11–13 to provide a transcriptomic reference, while the MERFISH dataset provides spatial context for a single PCW-12 heart at near-cellular resolution across the whole organ. Per-cell quality metrics (UMI counts, genes detected, mitochondrial fraction, cell volume) and standard metadata (sex, donor, cell-cycle phase, FOV, section) are pre-computed in the AnnData `obs` fields and used directly in the figures below.

2 scRNA-seq quality and composition

2.1 Per-cell QC metrics

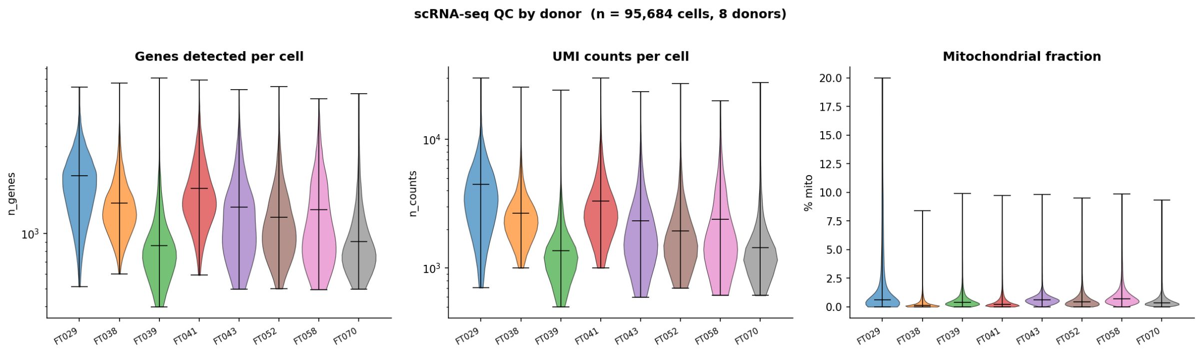

Figure 1. Per-cell QC distributions for the scRNA-seq atlas, split by donor ($n = 8$ donors). Median values across all 95,684 cells: **1,375** genes per cell, **2,389** UMI counts per cell, and **0.41%** mitochondrial content. Donor FT029 has a longer mitochondrial tail, but the bulk of cells in every donor sit below 5% — well within accepted thresholds for fetal tissue scRNA-seq. The log-scaled n_counts panel shows comparable depth across donors except for FT039 and FT070, which are slightly shallower; this is consistent with their smaller cell counts.

2.2 Cell-type composition

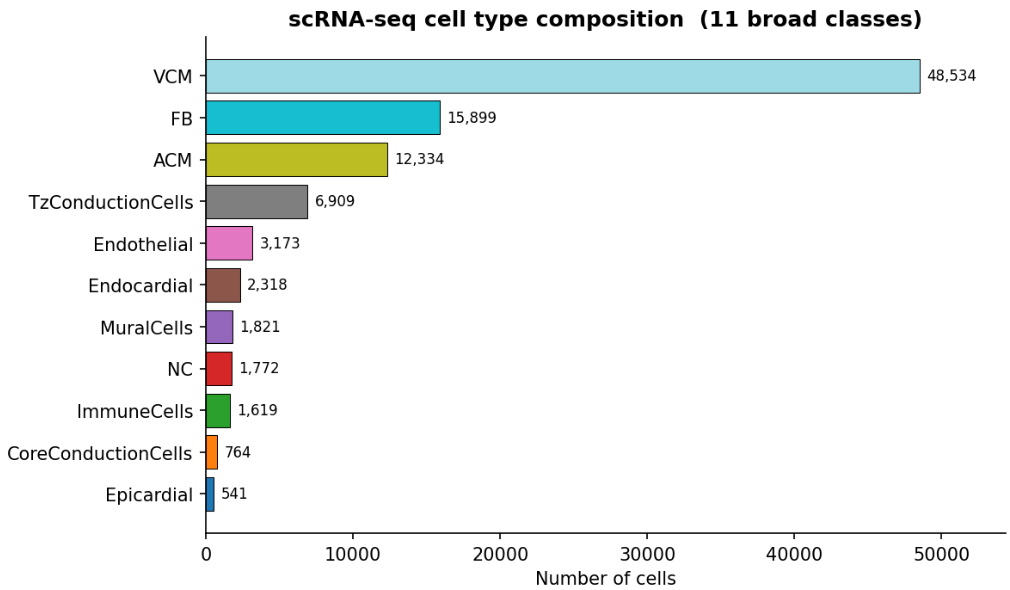

Figure 2. Cell counts per broad cell-type label (celltype_label, 11 classes). Ventricular cardiomyocytes (**VCM**, 50.7%) dominate the atlas, followed by fibroblasts (**FB**, 16.6%) and atrial cardiomyocytes (**ACM**, 12.9%). Conduction-system, endothelial, endocardial, mural, neural-crest, immune and epicardial populations together account for the remaining 19.8% — an expected composition for a fetal cardiac atlas. A fine-grained label (celltype, 111 classes) is also available for more granular analyses.

2.3 Donors, developmental stage, sex, and cell cycle

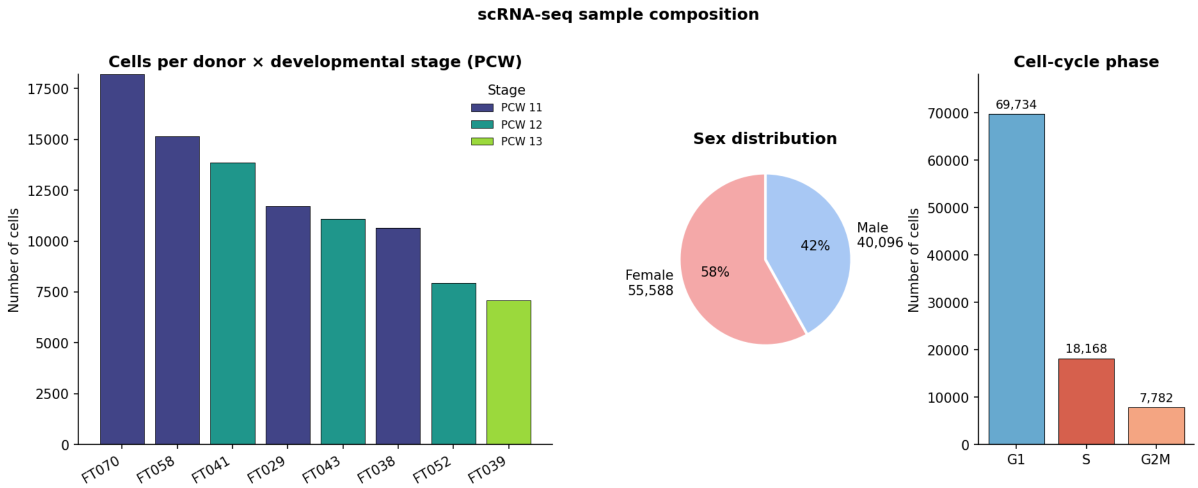

Figure 3. Sample composition. **Left:** cells per donor coloured by developmental stage (PCW). Each donor was sampled at a single PCW: FT029, FT041, FT043, FT052 at PCW 12; FT038, FT058, FT070 at PCW 11; FT039 at PCW 13. PCW-11 cells dominate the atlas (55,730 cells, 58%) followed by PCW 12 (32,870, 34%) and PCW 13 (7,084, 7%). **Centre:** sex distribution (58% female, 42% male). **Right:** cell-cycle phase scoring — ∼27% of cells are in S or G2/M, consistent with the high proliferative activity expected during mid-gestation cardiogenesis.

3 MERFISH quality and composition

3.1 Per-cell QC metrics

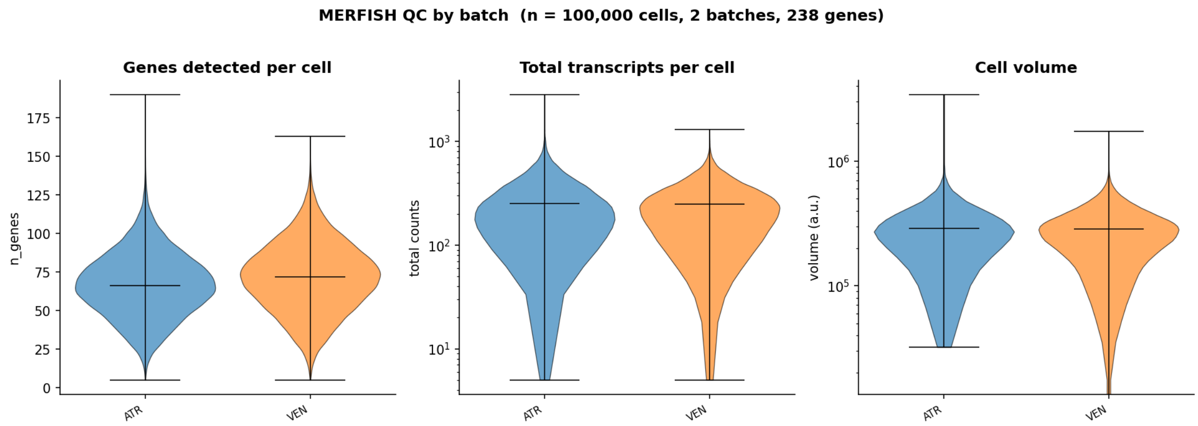

Figure 4. MERFISH per-cell QC, split by batch (ATR = atrial-region sections, VEN = ventricular-region sections). Median values: **69** genes detected per cell out of the 238-gene panel, **251** total transcripts per cell, and a cell volume of $\mathbf{2.86{\times}10^5}$ a.u. The two batches have nearly identical distributions across all three metrics, indicating that batch effects on per-cell yield are minimal. The log-scale total-counts panel reveals a small low-yield tail in both batches that may benefit from filtering for sensitive downstream analyses.

3.2 Cell-type composition

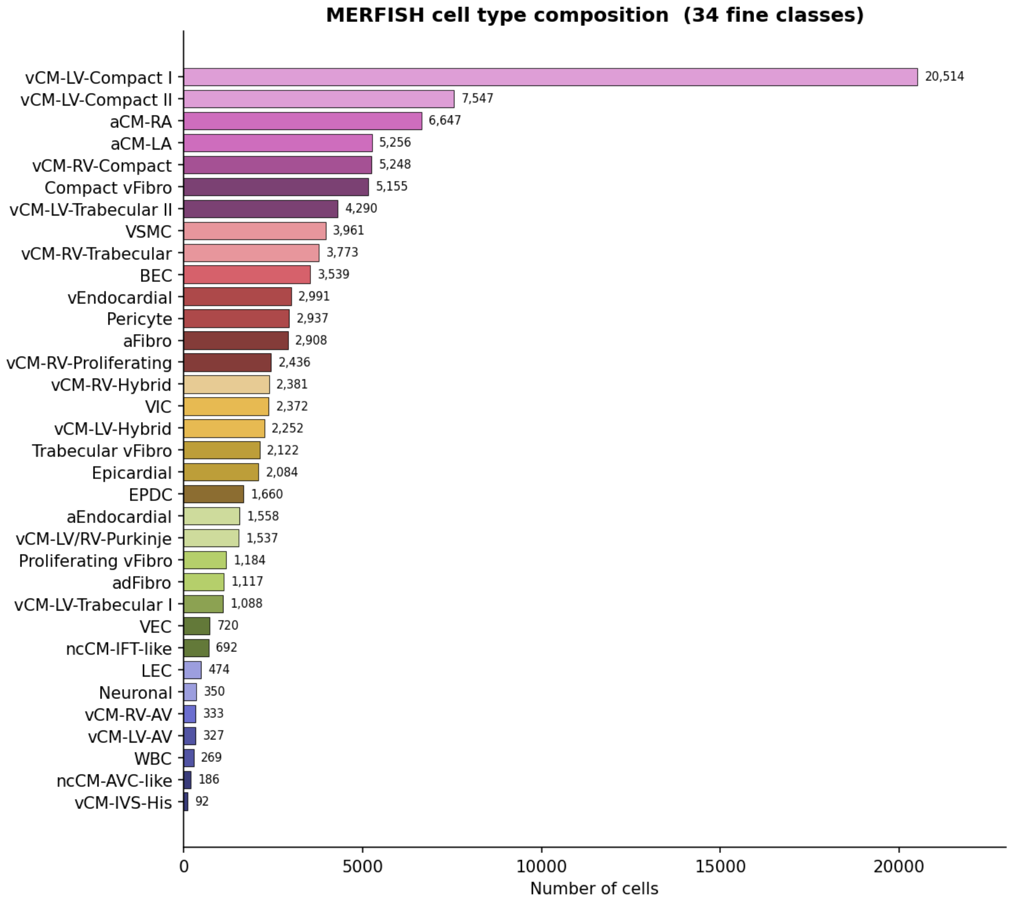

Figure 5. MERFISH cell-type composition (34 fine classes, ranked by abundance). The most abundant population is vCM-LV-Compact I (20,514 cells, 20.5%), reflecting the size of the compact left ventricular myocardium at PCW 12. Most rare populations (WBC, ncCM-AVC-like, vCM-IVS-His) still have ≥90 cells, providing sufficient sample sizes for cell-type-level spatial analysis.

3.3 Sectioning and field-of-view structure

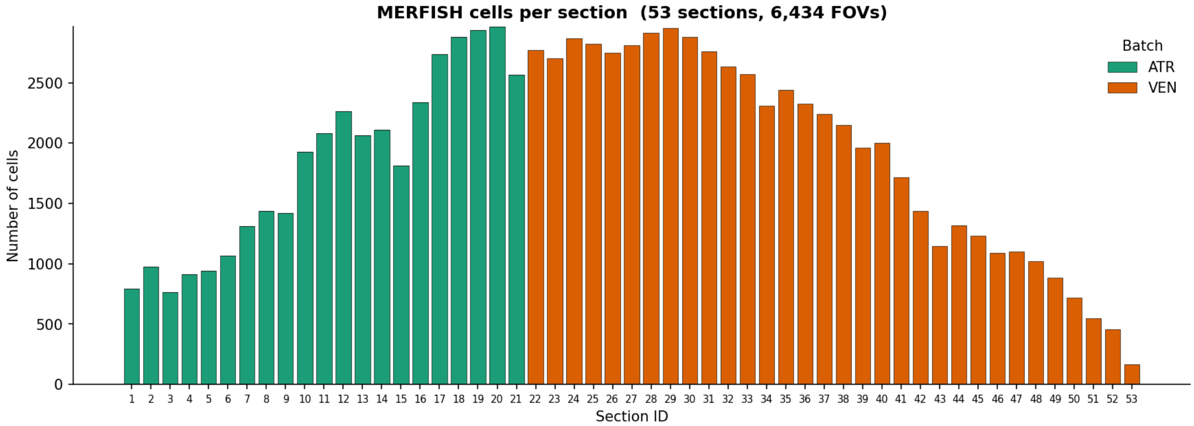

Figure 6. Cells per serial section (53 sections total, organized into 6,434 fields of view across two batches). Sections 1–21 (**ATR**, green) cover atrial regions; sections 22–53 (**VEN**, orange) cover the ventricles. The roughly bell-shaped yield within each batch reflects the varying physical cross-section of the developing heart along the axial direction — central sections capture the largest tissue area and therefore the most cells.

3.4 2D anatomical layout

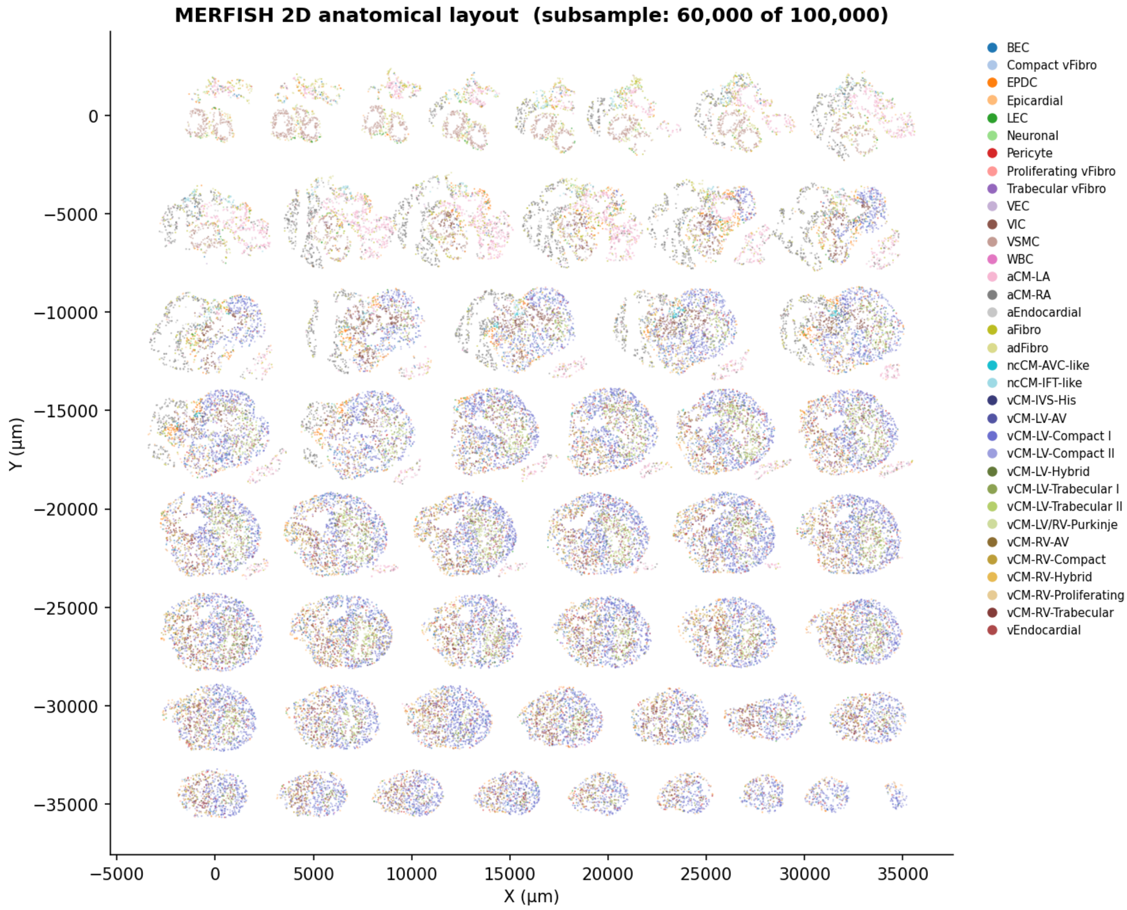

Figure 7. 2D arrangement of all 53 serial sections in the X_spatial embedding, coloured by the 34-class fine cell type. Each "island" is one anatomical section. The atrial sections (top rows) are smaller and morphologically distinct from the ventricular sections (bottom rows), with progressively larger cross-sections through the body of the heart. Within each section, spatial structure is clearly visible: ventricular cardiomyocyte populations (purple/blue tones) form the main chamber walls; epicardial/fibroblast populations (orange/yellow) decorate the outer rim; endothelial and conduction populations occupy distinct focal regions. Subsampled to 60,000 cells for legibility.

3.5 3D anatomical view

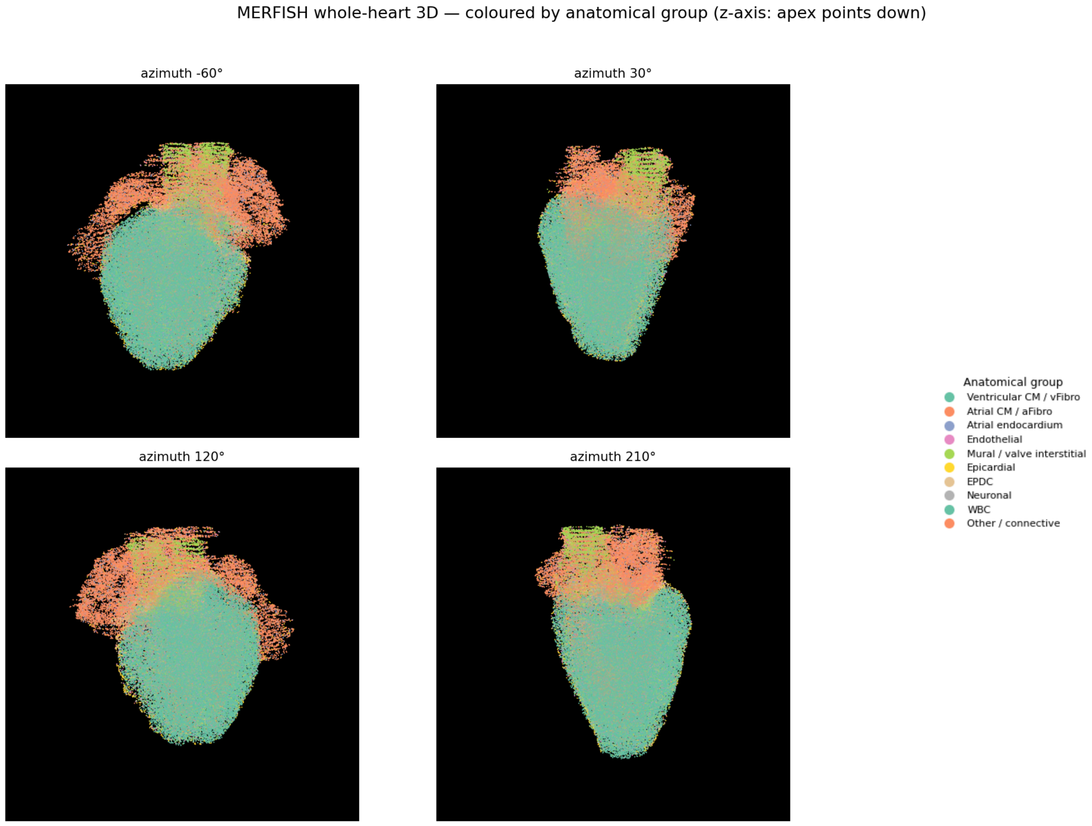

Figure 8. 3D rendering of all 100,000 MERFISH cells using the X_spateo_update embedding (z-axis flipped so the apex points down), coloured by 10 broad anatomical groups. Four azimuth viewpoints $(-60°, 30°, 120°, 210°)$ span the long axis. The atrial cardiomyocytes (orange) cap the upper pole, the ventricular cardiomyocytes (teal) form the bulk of the lower body, and the valve interstitial / mural cell layer (green) demarcates the atrio-ventricular plane. Fibroblasts and endothelial populations form a thin envelope on the outer surface and along the chamber linings.

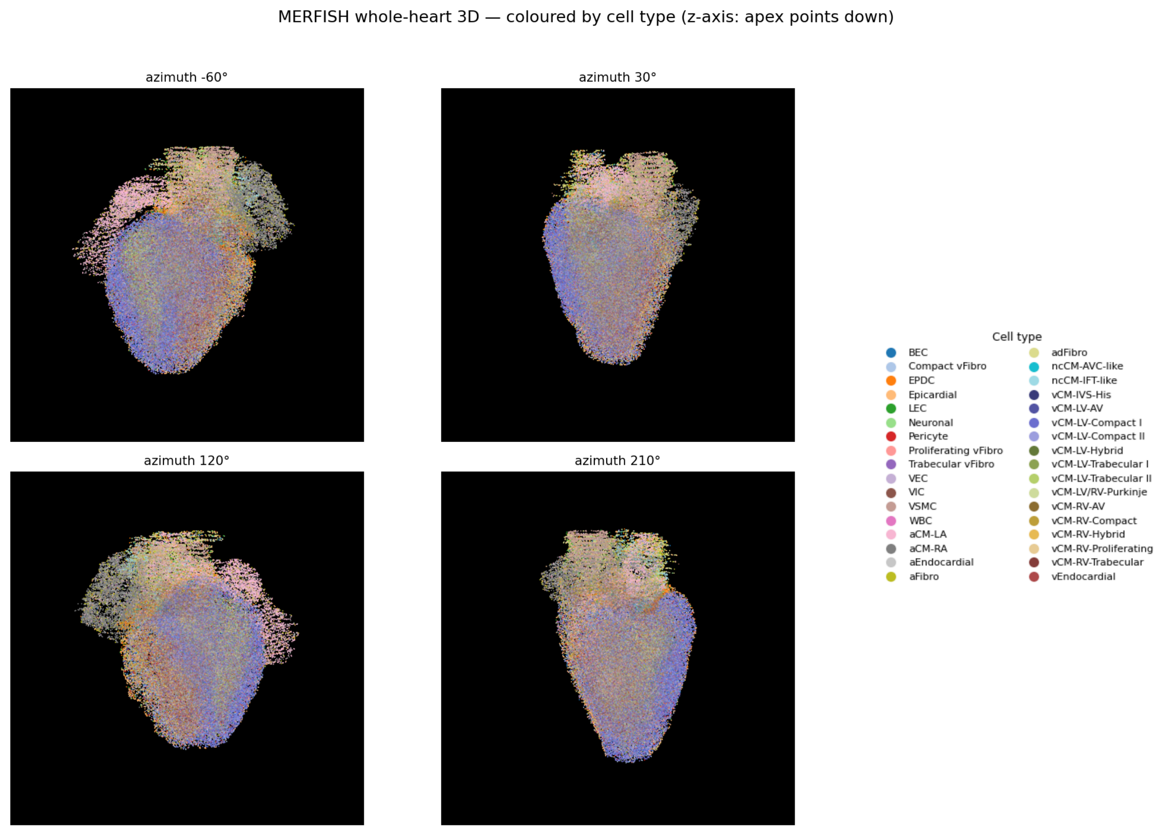

Figure 9. Same 3D rendering as Fig. 8 but coloured by the full 34-class fine annotation. Atrial subtypes aCM-LA and aCM-RA are visible at the top; the ventricular wall is dominated by vCM-LV-Compact and vCM-LV-Trabecular, with right-ventricular and septal populations on the opposing side; conduction-related populations (vCM-LV/RV-Purkinje, ncCM-AVC-like, ncCM-IFT-like) occupy focal regions consistent with their expected anatomical niches. Animated rotations are also available (`merfish_3d_chamber.gif`, `merfish_3d_celltype.gif`).

4 Cross-modality comparison

4.1 Broad cell-type composition

To compare the two modalities on equal footing, the 34 fine MERFISH cell types were collapsed onto the 11-class broad annotation used by the scRNA-seq atlas (e.g., `vCM-LV-Compact I`, `vCM-RV-Compact`, … → `VCM`; `Compact vFibro`, `aFibro`, … → `FB`). Nine broad classes are shared between the two datasets: VCM, ACM, FB, MuralCells, Endothelial, Endocardial, Epicardial, NC, ImmuneCells. Two label families do *not* match exactly: scRNA-seq distinguishes Tz- from Core-conduction cells, while MERFISH groups conduction populations differently (vCM-LV/RV-Purkinje, ncCM-IFT-like, ncCM-AVC-like) — shown as grey labels in Fig. 10.

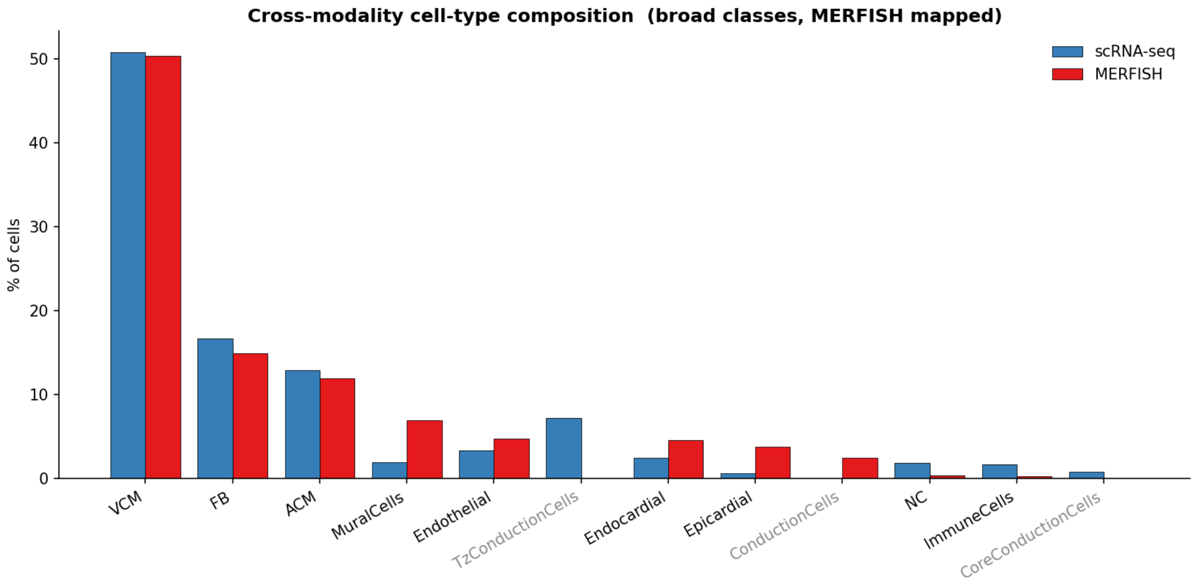

Figure 10. Per-modality cell-type fractions for the broad classes. The two modalities agree closely on the dominant populations: **VCM** is ∼50% in both, **FB** 16.6% vs. 14.9%, and **ACM** 12.9% vs. 11.9%. MERFISH over-represents some endothelial and mural populations relative to scRNA-seq — this is expected because spatial methods have less drop-out for sparse stromal populations than droplet scRNA-seq. Greyed labels indicate classes that do not have an exact counterpart in the other modality due to annotation-scheme differences.

4.2 Joint UMAP

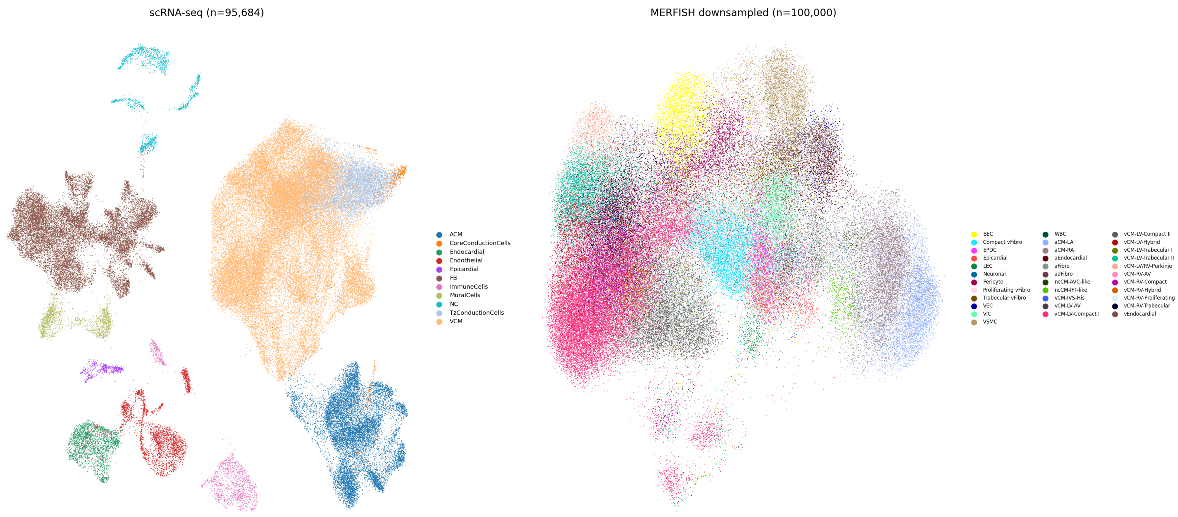

Figure 11. Side-by-side UMAPs. **Left:** scRNA-seq UMAP (95,684 cells) computed from the top 3,000 highly variable genes ($\rightarrow$ PCA(50) $\rightarrow$ kNN(15) $\rightarrow$ UMAP), coloured by the 11 broad classes. **Right:** the precomputed MERFISH UMAP (100,000 cells, X_umap) coloured by the 34 fine classes. Both atlases show the expected dominance of cardiomyocyte clusters with smaller, well-separated stromal, endothelial, and immune islands.

5 Summary and downstream readiness

Metric	**scRNA-seq**	**MERFISH**
Cells	95,684	100,000
Genes / panel	31,019	238
Median genes / cell	1,375	69
Median counts / cell	2,389	251
Median % mitochondrial	0.41%	—
Median cell volume	—	2.86×10^5 a.u.
Donors / specimens	8 / 24 samples	1 / 53 sections
PCW span	11, 12, 13	12
Cell-type classes	11 broad / 111 fine	34 fine

Table 2. Per-modality summary statistics.

The two atlases are complementary: scRNA-seq provides full-transcriptome coverage across multiple donors and developmental stages, while MERFISH provides spatial coordinates and cell-type localization within a single PCW-12 heart. Quality metrics are within standard ranges for both technologies, batch effects (ATR vs. VEN in MERFISH; donor in scRNA-seq) are modest, and the broad cell-type compositions agree closely. The nine shared broad classes (VCM, ACM, FB, MuralCells, Endothelial, Endocardial, Epicardial, NC, ImmuneCells) form the natural anchor for joint analyses such as MOSCOT optimal-transport mapping (see `REPORT.md` for the disease-gene spatial-imputation pipeline that builds on these data).

Reproducibility. Figures generated by `scripts/05_data_overview.py`; metadata snapshot at `figures/overview/_metadata.json`; summary statistics at `figures/overview/_summary_stats.json`.